Article

fp-tools: A Reproducible Platform for ATAC-seq Footprinting and Regulatory Motif Analysis

Yaoxiang Li [1] and Chunling Yi [1,*]

[1] Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Georgetown University, Washington, DC, USA; yl814@georgetown.edu (Y.L.); cy232@georgetown.edu (C.Y.)
* Correspondence: cy232@georgetown.edu (C.Y.)

Abstract

ATAC-seq footprinting enables near-base-resolution inference of transcription-factor (TF) occupancy across the genome. However, the field lacks a computationally efficient, user-friendly, and integrated platform that accepts both bulk and single-cell ATAC-seq data, supports both de novo motif discovery and matching to public motif databases, and generates interactive and publication-ready visual outputs. Here, we present `fp-tools`, a Python package that builds on the widely used ATAC-seq footprinting framework TOBIAS. We substantially improve computational efficiency across key steps, including Tn5 bias correction, footprint calling, and motif matching. In addition, `fp-tools` introduces de novo motif discovery, replicate-aware differential footprinting, and aggregation analysis, features not provided as native TOBIAS workflows. We further streamline a single-cell ATAC-seq footprinting workflow, provide interactive visualization of analysis results, and implement a graphical user interface (GUI) that enables code-free execution of all package commands. Using public bulk ATAC-seq replicates and single-cell datasets, we demonstrate that `fp-tools` quantitatively recovers cell-type-specific TF-binding patterns and identifies de novo binding motifs that are not captured by existing public motif databases. Together, `fp-tools` provides an efficient, accessible, and comprehensive platform for TF footprint analysis across bulk and single-cell ATAC-seq studies.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; pseudobulk ATAC-seq; replicate-aware reporting; reproducibility; command-line bioinformatics

1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of active regulatory DNA [1]. Within accessible regions, sequence-specific transcription factors (TFs) can protect bound DNA from transposition, producing local cut-site depletions, or "footprints," with increased accessibility in nearby flanks. Detecting these motif-centered cut-site patterns supports inference of TF occupancy, differential regulatory activity between conditions, and interpretation of non-coding regulatory variants. Earlier footprinting and occupancy models, including CENTIPEDE, Wellington, PIQ, and DeFCoM, established the value of combining sequence information with chromatin accessibility profile shape for TF-binding-site inference [2–5].

Despite this progress, practical ATAC-seq footprinting remains difficult to run as a complete and reproducible analysis. Widely used tools such as TOBIAS [6] and HINT-ATAC [7] provide enzymatic bias correction and footprint scoring for bulk ATAC-seq data.

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Multiscale and nucleosome-aware approaches, including the PRINT/scPrinter family [8], extend footprint modeling across different scales of chromatin protection. Supervised TF-binding-site predictors such as maxATAC [9] and sequence-to-profile models such as ChromBPNet [10] provide complementary prediction frameworks. Motif-discovery and motif-comparison tools, including MEME/STREME [11,12] and Tomtom [13], are commonly used after peak or footprint selection, and curated databases such as JASPAR [14] remain central for known-motif matching. These tools are valuable, but they are typically assembled through separate command interfaces, file conventions, report formats, and visualization steps.

This fragmentation creates a gap for applied users. A complete analysis often requires bulk or single-cell ATAC-seq input handling, Tn5 sequence-bias correction, footprint calling, motif matching against public databases, optional de novo motif discovery, differential footprint analysis, aggregate visualization, and publication-ready reporting. Single-cell ATAC-seq adds another layer of complexity because sparse per-cell fragments must usually be aggregated into interpretable cell-type or neighborhood-level signals before conventional footprinting can be applied. For many biology laboratories, the limiting step is not the absence of any individual algorithm, but the lack of a single efficient and accessible platform that connects these steps while preserving the analysis record.

We developed `fp-tools` to fill this workflow gap. `fp-tools` builds on the widely used TOBIAS ATAC-seq footprinting framework while improving the surrounding platform for speed, accessibility, and end-to-end analysis. The package keeps the interpretable motif-centered footprint model familiar to TOBIAS users, but implements faster core operations for Tn5 bias correction, footprint calling, and motif matching, and adds workflow components that are not provided as native TOBIAS workflows. These include de novo motif-discovery preparation and matching, replicate-aware differential footprint reports, motif-centered aggregation plots, pseudobulk single-cell ATAC-seq support, interactive HTML reports, and a graphical user interface for code-free execution of the same command-compatible workflows.

Here, we describe the `fp-tools` implementation and evaluate it using public bulk ATAC-seq replicates and public single-cell ATAC-seq datasets. The examples test whether the platform produces reproducible corrected signals, footprint scores, differential motif reports, de novo motif summaries, and cell-type-specific pseudobulk footprint signatures. We position `fp-tools` as an integrated and user-accessible extension of established ATAC-seq footprinting practice: it does not replace specialized single-cell, deep-learning, or motif-discovery tools, but connects common TF-footprinting tasks into a single auditable workflow that can be run from the command line, batch YAML files, or the GUI.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

`fp-tools` implements performance-critical signal operations with Cython-backed routines. The workflow has four method layers: Tn5 sequence-bias correction, footprint scoring over corrected signal, motif-anchored differential TF-binding analysis, and aggregate visualization around TFBS or region sets. These layers share common region, signal, motif, and sample-label conventions so intermediate files can be passed through a complete analysis without format-specific glue code. YAML configuration files provide a reproducible way to save and rerun the same analysis without changing the underlying methods.

Following the signal model popularized by TOBIAS [6], `fp-tools` treats base-resolution, bias-corrected cut-site profiles as the primary footprinting signal. This design deliberately preserves the familiar bulk ATAC footprinting interpretation: the biological unit is a corrected center-versus-flank cut-site contrast at motif or candidate sites, while the software layer adds consistent command names, manifests, reusable configuration, replicate summaries, and report generation around that signal. For footprint calling, let x_i denote corrected Tn5 cut-site signal around candidate position p. The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|}\left(\sum_{i \in L} x_i + \sum_{i \in R} x_i\right) - \frac{1}{|C|}\sum_{i \in C} x_i,$$

where C is the protected center and L and R are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

2.2. De Novo Motif Discovery Preparation

The de novo motif-discovery workflow begins from candidate footprint intervals, which may be produced by footprint calling or supplied by the user. For each candidate interval j with center p_j, strand a_j, and flank size h, the workflow exports a candidate-centered sequence

$$S_j = G[p_j - h, p_j + h],$$

reverse-complementing the sequence when strand information indicates the negative strand. The workflow records the input intervals, genome build, flank width, software versions, and MEME, DREME, STREME [11,12,15] or optional Tomtom [13] settings needed to reproduce each run. STREME was designed for accurate and versatile motif discovery from enriched sequence sets [12]. The same entry point can be used in two modes. In de novo-only mode, discovered motifs are reported as an independent motif set. In database-plus-de-novo mode, discovered motifs are matched to a known database such as JASPAR 2026 [14] and can be appended to the known motif set for downstream differential footprint analysis. This extends the standard TOBIAS-style motif analysis path, which relies on supplied motif collections, by making candidate FASTA export, motif discovery, motif comparison, and database augmentation a reproducible companion workflow rather than an external manual step.

2.3. Pseudobulk Fragment Aggregation

The pseudobulk workflow groups single-cell ATAC fragments by a metadata column, such as cell type, cluster, donor, or treatment. Let F_b be the multiset of fragments assigned to barcode b, and let $g(b)$ be its group label. The pseudobulk fragment collection for group k is the concatenated multiset

$$F_k = \mathrm{concat}_{b:g(b)=k}(F_b),$$

so repeated fragments and fragment multiplicity are preserved. Groups that pass user-defined cell-count and fragment-count filters are written as bulk-like fragment files with manifest records. When genome sizes are supplied, `fp-tools` can also emit CPM-normalized cut-site bigWigs,

$$y_{k,t} = 10^6 \frac{c_{k,t}}{\sum_u c_{k,u}},$$

where $c_{k,t}$ is the cut-site count for group k at genomic position t. These tracks can be used directly by the same motif-centered aggregate plotting workflow used for bulk ATAC-seq data. Thus, users can run a TOBIAS-like bulk footprinting analysis on single-cell data without first constructing each pseudobulk sample outside the footprinting workflow. When requested, the full route writes pseudo-paired BAMs from grouped fragments and then runs correction, scoring, motif-aware comparison, and plotting with the same conventions used for bulk ATAC-seq.

For visualization of cell-level marker footprint signatures, `fp-tools` also supports a K-nearest-neighbor (KNN) smoothing layer over single-cell fragments. For each marker TF, exact motif centers are scanned inside the selected accessible-bin universe. Fragment cut sites are counted in a ± 100 bp window around those motif centers for each barcode. Each cell is then assigned a local neighborhood profile by summing its 75 nearest neighbors in the single-cell embedding, and the same flank-minus-center footprint-signature score is computed from that neighborhood profile. Scores are oriented so that the expected marker cell type has the positive direction and then standardized per TF for UMAP display. These KNN-smoothed signatures are used only as per-cell visualization scores; formal corrected footprint inference remains based on the grouped pseudobulk ATAC-correction and footprint-scoring workflow.

2.4. Replicate-aware Differential Footprint Analysis and Quantile Normalization

Replicate-aware analysis is integrated directly into the differential footprint comparison workflow. The differential comparison retains conventional motif-centered footprint-score logic while adding repeated condition labels, replicate-level summaries, optional quantile normalization, and two complementary testing views. Repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, define biological replicate groups while preserving the original differential-footprint output structure. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c.

The default distribution-comparison view is used directly by `diff-footprints`. It supports either one sample per group or multiple samples per group. When repeated condition names are supplied, replicate scores are summarized into condition-level means and SDs before motif-level comparisons are reported; when only one sample is present per group, the same motif and background-distribution comparison is still available, but replicate-derived SD and SE summaries are not interpreted as replicate uncertainty. A second empirical-Bayes replicate-matrix view is available when per-replicate motif-score matrices are built for a comparison. This view is limma-like rather than a direct dependency on the R `limma` package: motif scores are transformed as $\log_2(1000x + 1)$, condition effects are estimated from replicate groups, and the per-motif pooled variance is moderated toward the median variance across motifs before computing a moderated t-statistic and Benjamini–Hochberg-adjusted q-values. This empirical-Bayes view is used as a replicate-aware audit and alternative reporting view for replicate-supported comparisons, while the native distribution-comparison columns are preserved for traceability. Relative to a conventional TOBIAS-style two-condition report, the added workflow-level contribution is explicit handling of repeated biological condition labels, side-by-side single-sample and replicate-supported comparison modes, and separate statistical and display-priority columns in the exported tables.

With `-normalization sample-quantile`, each replicate is normalized before condition averaging. The normalization fit uses finite positive scores only, because negative or zero footprint-score values do not define a stable multiplicative quantile ratio. For sample s, `fp-tools` computes empirical score quantiles $q_s(p)$ on a 0.05–0.99 quantile grid and the reference quantile $q_*(p) = S^{-1} \sum_s q_s(p)$ across the S input samples. It then smooths the ratio $q_*(p)/q_s(p)$ and fits a sigmoid multiplicative factor $g_s(x)$; if the sample has too few positive values or the curve fit is ill-conditioned, a constant positive factor is used instead. During application, scores outside the fitted positive-score range use the nearest fitted range boundary for the normalization factor. Each score is transformed as

$$\tilde{x}_{s,i} = \max\{0,\, x_{s,i} g_s(x_{s,i})\}.$$

The final maximum operation clips transformed values at zero, preserving the interpretation that positive normalized values represent footprint-like center-versus-flank evidence. The `condition-quantile` mode first averages replicate background distributions within each condition, fits one factor $g_c(x)$ per condition, and applies that condition-level factor to all replicates in the condition. `none` disables quantile normalization. The same normalization helper is used by aggregate plotting, so differential comparison and motif-centered visualization are normalized consistently.

After normalization, condition-level scores are summarized as

$$\bar{x}_{c,i} = \frac{1}{n_c} \sum_{r=1}^{n_c} \tilde{x}_{c,r,i}, \qquad s_{c,i} = \sqrt{\frac{1}{n_c - 1} \sum_{r=1}^{n_c} (\tilde{x}_{c,r,i} - \bar{x}_{c,i})^2}.$$

For a comparison between conditions c_1 and c_2, `fp-tools` reports a footprint-score difference and a stabilized log fold change,

$$\Delta_i = \bar{x}_{c_1,i} - \bar{x}_{c_2,i}, \qquad L_i = \log_2 \frac{\bar{x}_{c_1,i} + \epsilon}{\bar{x}_{c_2,i} + \epsilon},$$

where ϵ is estimated from the background signal unless supplied explicitly. When both conditions have replicate support, approximate standard errors are reported as descriptive summaries. Here, μ_1 and μ_2 denote the condition-level mean scores for the two conditions at the included motif sites, s_1 and s_2 denote the corresponding empirical SD estimates across replicates, and n_1 and n_2 are replicate counts:

$$\mathrm{SE}(\Delta) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

$$\mathrm{SE}(L) \approx \frac{1}{\ln 2} \sqrt{\frac{s_1^2}{n_1(\mu_1 + \epsilon)^2} + \frac{s_2^2}{n_2(\mu_2 + \epsilon)^2}}.$$

In the distribution-comparison view, the reported `change` and `pvalue` columns are computed at the motif-reporting level. Candidate sites are included when either condition has a positive motif-site score, site-level log fold changes are averaged within peak intervals, and the observed motif statistic is defined as the difference between the fitted observed mean log fold change and the fitted background mean, scaled by the average of the observed and background SD estimates. `fp-tools` then simulates background change statistics from the fitted background distribution using the same number of observed intervals and applies a one-sample t-test comparing the simulated background changes with the observed motif statistic. P-values are adjusted across motifs within each contrast using Benjamini–Hochberg, yielding the `qvalue_bh` column and `significant_fdr05` flag. These p-values are screening statistics for the motif-centered report, especially at small replicate counts; the SE columns are reported separately as descriptive uncertainty summaries and are not themselves the hypothesis-test statistic. The `highlighted` column is retained only as a report-selection and visualization flag, not as a formal multiple-testing-corrected significance label. Additional replicate-oriented columns include replicate counts, condition score SDs, mean Δ, mean L, and standard-error summaries.

2.5. Public ATAC Processing and Reproducibility Provenance

The bulk replicate demonstration uses true biological replicates from the Buenrostro et al. ATAC-seq study [1]: GM12878/B-cell samples SRR891268 and SRR891269, and day-1 CD4 T-cell samples SRR891275 and SRR891276. Reads were trimmed with fastp [16], aligned with Bowtie2 [17], processed with SAMtools [18] and BEDTools [19], and peaks were called with MACS3 using the MACS model-based peak-calling framework [20]. Problematic genomic regions were filtered using the ENCODE blacklist [21]. Replicate peaks were merged into a common peak universe, bias-corrected, footprint-called, and analyzed by differential footprint comparison with and without sample-level quantile normalization. The analysis used JASPAR 2026 CORE vertebrates non-redundant motifs [14] for known-motif scanning. Exact command-line parameters and software versions are recorded in source TSV files in the manuscript `tables` directory and summarized in Tables S1 and S2.

2.6. Public Data, Demonstration Tiers, and Metrics

Public inputs are organized as versioned TSV manifests recording source, demonstration tier, cell type, donor, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. Discovery and download scripts query public repositories such as ENCODE [22] for released human GRCh38 ATAC-seq records, write resumable download reports, and keep raw public data out of version control. The tracked repository contains the scripts, manifests, schemas, small reference inputs, source tables, and figure code needed to rerun the manuscript analyses.

The demonstration design follows published recommendations for transparent and reproducible computational method evaluation [23,24], but it is not intended as an exhaustive head-to-head benchmark of all ATAC footprinting methods. Instead, it is organized as a tiered software-evaluation design (Table 1): a small-reference regression tier; a bulk ATAC replicate tier; a depth-robustness tier over downsampled ATAC; a de novo motif-preparation tier; and a real-data pseudobulk tier. Summary tables report command success, output integrity, runtime, memory, differential footprint summaries, motif-recovery summaries, and generated figures for the analyses included in this manuscript.

Table 1. Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and compact reference results are committed.

Tier	Purpose	Primary outputs
Small reference	Installation and regression checks on compact inputs	command success, output existence, stable summaries
Bulk ATAC replicates	Public bulk ATAC replicate demonstrations for footprinting and differential reports	corrected tracks, footprint scores, motif reports, aggregate plots
Depth robustness	Shallow-depth sensitivity (50/25/10/5%)	downsampled workflow summaries and output checks
De novo motif prep	Candidate FASTA and motif-discovery validation	exported FASTA files and motif summaries
Pseudobulk	Single-cell extension and real-data demonstration	10x PBMC pseudobulk fragments, corrected footprint tracks, pairwise TF differential plots, KNN marker-signature UMAPs

2.7. Reproducibility Engineering

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint calling, differential footprint analysis, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using compact reference inputs, builds source and wheel artifacts, and verifies the installed console scripts. The repository includes `environment.yml`, a Dockerfile, release metadata, and Make targets for test, build, and manuscript-smoke workflows; these choices follow FAIR-oriented research-software principles and established bioinformatics environment practices [25–27]. Detailed reviewer instructions are kept in `docs/reproduce-paper.md`. Benchmark scripts record random seeds, tool versions, command lines, and resolved manifests alongside the corresponding outputs. Figure source tables and editable figure files are retained for auditability.

3. Results

We evaluate `fp-tools` as a software and reproducibility platform along three axes: workflow integration and command stability, controlled local engineering measurements, and biological plausibility in public bulk and single-cell ATAC-seq demonstrations.

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the core ATAC-seq footprinting steps into a single, consistent workflow with shared region/signal conventions, so a complete analysis, including bias correction, footprint calling, differential footprint analysis, and aggregate visualization, runs without crossing package boundaries or reformatting intermediates. YAML configuration records the same analysis in a machine-readable form for reruns and batch processing (Figure 1).

3.2. Interactive Interfaces and Report Review

In addition to direct command-line execution, `fp-tools` provides browser-based interfaces for configuring analyses and reviewing results. The GUI writes reusable YAML configurations that remain compatible with command-line execution and provides guided panels for the same core commands shown in the command-line documentation (Figure 6). Differential-footprint and aggregate-batch workflows also produce self-contained HTML reports with embedded compressed data, searchable motif controls, editable group colors, synchronized aggregate profiles, and SVG export for publication-ready editing.

3.3. Regression-tested Workflow Outputs

The footprinting workflow is covered by deterministic regression tests that pin summary outputs for footprint calling, differential footprint analysis, and aggregate plotting. Public entry-point checks and YAML dry-runs verify that the command surface remains stable. These tests provide a reproducible contract for core analyses while allowing integrated modules to share the same input and output conventions.

3.4. De Novo Motif Discovery Complements Database-based Motif Analysis

The de novo motif-preparation path exports candidate-centered FASTA files, applies MEME, DREME, or STREME, optionally adds a Tomtom comparison against a known motif database, and summarizes motif results into TSV/HTML reports with inline consensus logos. This keeps motif discovery reproducible while relying on established motif-discovery algorithms. The workflow can be used alone to discover recurring patterns in candidate intervals, or after known-motif scanning to find additional motifs not represented in the selected database. In relation to TOBIAS, this section evaluates an added motif-discovery handoff around the footprinting workflow, not a new center-versus-flank footprint scoring model.

We therefore evaluated whether candidate-derived motifs add information beyond a fixed database using a 3-vs-3 public ENCODE K562 versus HepG2 ATAC-seq comparison [22]. Candidate footprints were called from bias-corrected cut-site tracks, condition-specific candidate FASTA files were exported with 75 bp flanks, and STREME was run reciprocally with K562 candidates against HepG2 controls and HepG2 candidates against K562 controls. These discovered motifs were then used in two modes. In de novo-only mode, the candidate-derived motifs were tested as their own motif set. In supplement mode, the same motifs were appended to a known motif database before running the sample-quantile-normalized differential-footprint analysis.

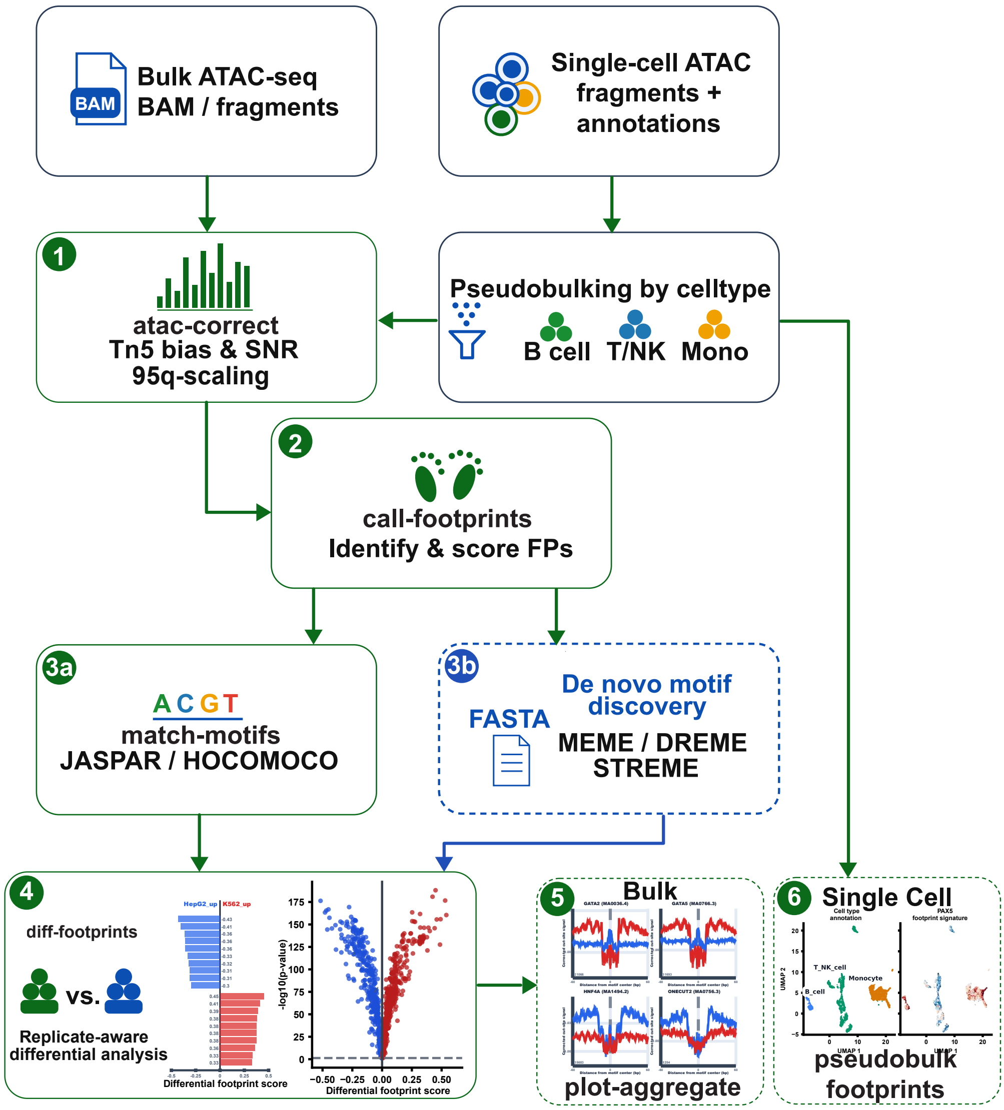

Figure 1. Overview of the `fp-tools` ATAC-seq footprinting platform. Bulk ATAC-seq BAM/fragments and single-cell ATAC fragments with annotations enter a shared command-oriented workflow. The workflow supports Tn5 bias correction, footprint calling, motif matching against curated databases, de novo motif discovery, replicate-aware differential footprint analysis, aggregate footprint visualization, and pseudobulk single-cell footprint analysis.

The full JASPAR 2026 vertebrate non-redundant set contained 1019 motifs, while the expanded reciprocal STREME runs contributed 500 candidate-derived motifs from both ENCODE cell lines. We therefore interpret this run as a site-recovery demonstration of how far de novo-only discovery can cover database-supported footprints, rather than as an exhaustive replacement for a curated motif database. In de novo-only mode, recovery was measured as base-pair overlap after merging the bound-site windows produced by the de novo scan and the corresponding JASPAR-supported scan. By this operational overlap metric, the discovered motifs recovered 84.0% of merged JASPAR-supported bound-site-window base pairs across the K562/HepG2 comparison, while 16.0% remained unique to the full JASPAR scan. The de novo-only scan also introduced additional candidate-derived bound-site windows, corresponding to 6.0% of the de novo-only merged window base pairs. In supplement mode, discovered motifs are appended to known motif sets before the same motif-aware differential footprint analysis. To create a sensitivity test, we also constructed a K562/HepG2-focused restricted JASPAR 2026 set with 867 motifs by removing common erythroid/K562 and liver/HepG2 motif families, then reran the comparison with and without the de novo motifs. In the interactive report, the restricted database yielded 92 display-priority motif families according to the `highlighted` report-selection flag; adding the 500 de novo motifs increased this browsing set to 153 families, including 24 de novo-derived motifs. Because `highlighted` is not a multiple-testing-corrected significance label, these counts are interpreted only as report-prioritization output, not as formal evidence for additional significant motif families. This sensitivity test illustrates the intended use case: when a chosen motif database is incomplete or conservative, candidate-derived motifs can be evaluated on their own and can also be carried forward into the same differential footprint report as database-supported motifs (Figure 2).

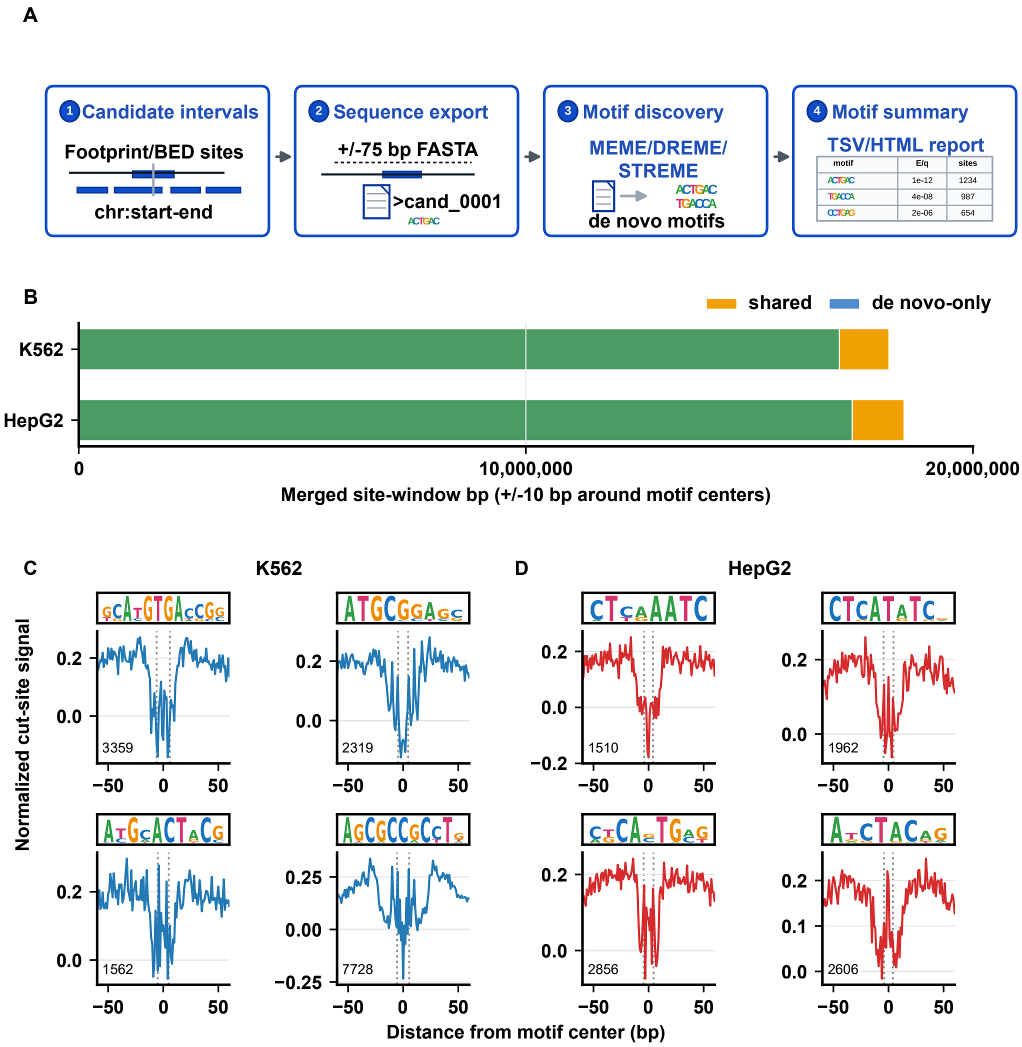

Figure 2. De novo motif discovery and validation workflow. (**A**) Candidate footprint or BED intervals are converted into centered FASTA sequences, analyzed with MEME/DREME/STREME, and summarized as motif tables and reports. (**B**) Merged site-window coverage compares shared and de novo-only motif-supported intervals in K562 and HepG2. (**C,D**) Representative K562 and HepG2 de novo motifs show footprint-like normalized cut-site aggregate profiles centered on motif instances.

A representative K562 versus HepG2 differential footprint report is shown in Figure 3. The report combines selected motif logos, top differential motif rankings, a volcano summary, and replicate-level aggregate footprint profiles in one browser-review view.

Figure 3. Interactive differential footprint report for K562 versus HepG2. The cropped browser report view summarizes selected differential motifs, motif logos, top differential footprint-score bar plots, a volcano plot, and replicate-level motif-centered aggregate profiles. Blue indicates HepG2-biased footprint scores and red indicates K562-biased footprint scores.

3.5. Replicate-aware Differential Footprint Analysis and Normalization

The replicate-aware workflow summarizes differential footprint results with effect sizes, replicate support, and uncertainty estimates while keeping the comparison anchored to a common peak and motif-site universe. In the Buenrostro example, B-cell and T-cell replicates are compared after motif scanning on the merged peak set, so no additional peak alignment step is performed at the reporting stage.

Replicate-aware differential footprint analysis adds an interpretable layer to condition comparison: repeated condition names define biological replicate groups, condition means are reported alongside individual replicate profiles, and sample-level quantile normalization applies the same distribution-matching step before replicate group averages are compared. The native distribution-comparison view supports both single-sample contrasts and multi-sample groups; replicate groups add condition SD, SE, and replicate-count summaries around the same motif-background comparison. For replicate-supported matrix analyses, the limma-like empirical-Bayes view provides a second report in which per-motif replicate variation is moderated across the motif set before p-values and q-values are assigned. The no-normalization mode retains the baseline comparison without an additional sample-level distribution match, whereas the sample-quantile mode applies distribution matching before condition-level comparison and aggregate plotting.

As a biological consistency check for the same 2-vs-2 Buenrostro replicate comparison, we inspected aggregate corrected cut-site profiles for a small illustrative set selected to show visible center depletion and known or plausible immune-cell relevance. BACH2 showed a B/GM12878-biased pattern in this dataset while remaining biologically non-exclusive because it is also reused in T-cell states. Conversely, JUNB and ATF7 showed stronger T-cell aggregate depletion; JUNB is an AP-1 family regulator of T-cell differentiation [28], and ATF7 is included here as a strong T-cell-deeper aggregate example rather than a lineage-specific marker. We also include IRF4 because TCR signaling can drive graded IRF4 expression in T cells [29], although its aggregate depletion is modest in this dataset. We treat these panels as an illustrative sanity check rather than a marker-classification benchmark; a more negative center-versus-flank profile indicates stronger footprint-like protection at motif centers.

3.6. Single-cell ATAC Fragments Support Pseudobulk Footprint Workflows

We evaluated pseudobulk aggregation using the public 10x Genomics PBMC granulocyte-depleted Multiome dataset [30]. The workflow used the official 10x ATAC fragments and GEX graph clusters, assigned 11,898 annotated cells to broad immune groups, and retained seven groups comprising 11,611 cells after requiring at least 300 cells and 50,000 fragments: B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and a mixed myeloid group. The prepared figure inputs used chromosomes chr1–chr22 plus chrX. The kept pseudobulks contained 13.6 to 144.0 million counted fragments per group and were written as indexed fragment files plus CPM-normalized cut-site bigWigs. The full `pseudobulk-footprints` route uses the same grouped fragments to emit pseudo-paired BAMs, run `atac-correct` with zero read shifting, score corrected footprint tracks, and optionally run motif-aware `diff-footprints` and aggregate plotting. Raw CPM cut-site tracks provide QC and chromatin-context signal alongside corrected footprint scores. These outputs were readable by the same aggregate-plotting code used for bulk ATAC data. This is the main practical extension beyond using TOBIAS directly on a prepared pseudobulk BAM or fragment track: `fp-tools` keeps barcode grouping, pseudobulk QC, correction, footprint scoring, motif-aware comparison, and aggregate plotting in one reproducible route. This pseudobulk route is complementary to established single-cell chromatin-accessibility ecosystems such as ArchR and Signac and to motif-activity baselines such as chromVAR [31–33].

The pseudobulk footprint result table was also used to prioritize marker TFs for cell-level visualization. We summarized the three broad-cell pairwise comparisons as volcano plots and labeled only predeclared marker TFs from the two cell types included in each comparison when the differential-footprint direction matched the expected lineage direction. Third-lineage markers and opposite-direction markers were not annotated.

We also applied the per-cell KNN visualization layer to the public 10x Genomics 5k PBMC single-cell ATAC dataset [34]. The demonstration used the Cell Ranger ATAC 2.0.0 fragment dataset, broad B-cell, monocyte, and T/NK annotations derived from the single-cell embedding, and motif-centered marker scores for PAX5, CEBPB, TCF7, and four additional high-change marker TFs from the pairwise pseudobulk screen: CEBPA, SPIB, ZBTB7B, and POU2F2. For each cell, the plotted value is a KNN-smoothed footprint-signature score, not an independently bias-corrected single-cell footprint call. We summarize these scores as a marker-by-cell-type heatmap and as UMAP projections for one representative marker from each broad cell type. This visualization highlights expected marker directions while avoiding overinterpretation of sparse per-cell cut-site counts: PAX5 and POU2F2 show B-cell-associated signal, CEBPB and CEBPA show monocyte-associated signal, and TCF7 and ZBTB7B show T/NK-associated signal; SPIB is included as an additional high-change B-cell marker from the pseudobulk screen (Figure 4).

3.7. Benchmark and Figure Reproducibility

The tiered demonstration design (Table 1), manifest builders, schema validator, engineering-runtime recorder, and figure-generation scripts provide an executable path from public inputs to publication-ready outputs. These resources serve three purposes. First, they verify that the ATAC-seq footprinting workflow produces stable outputs on small fixtures and public-data subsets. Second, they support the manuscript demonstrations for bulk replicate processing, de novo motif preparation, pseudobulk aggregation, replicate-aware normalization, and report generation. Third, they keep public inputs, parameters, and generated figure sources auditable as the workflow is rerun after code changes.

The benchmark pipeline writes resolved manifests, software versions, parameters, and plotting tables next to generated figures. This makes the figures auditable: a reader can inspect which samples, motifs, regions, and signal tracks were used, and the same tier can be rerun after code changes. Representative local engineering measurements summarize runtime and peak memory for one local workflow comparison using the same input bundle and command-level outputs (Figure 5). The browser GUI emphasizes the same command-first workflow while making example loading, YAML review, run history, and report inspection accessible to users who do not routinely work at the command line (Figure 6).

4. Discussion

`fp-tools` contributes a reproducible platform for ATAC-seq footprinting and motif-centered regulatory analysis. Its comparative strengths are interface consistency across bias correction, footprint calling, differential footprint analysis, and aggregate visualization; deterministic regression coverage; explicit handling of replicate-aware summaries; and integrated support for de novo motif preparation and pseudobulk aggregation using the same region, motif, and signal conventions (Table 2). Together, these properties address the fragmentation that often separates footprint scoring, motif interpretation, and report generation. The closest comparison is TOBIAS, which established a widely used integrated

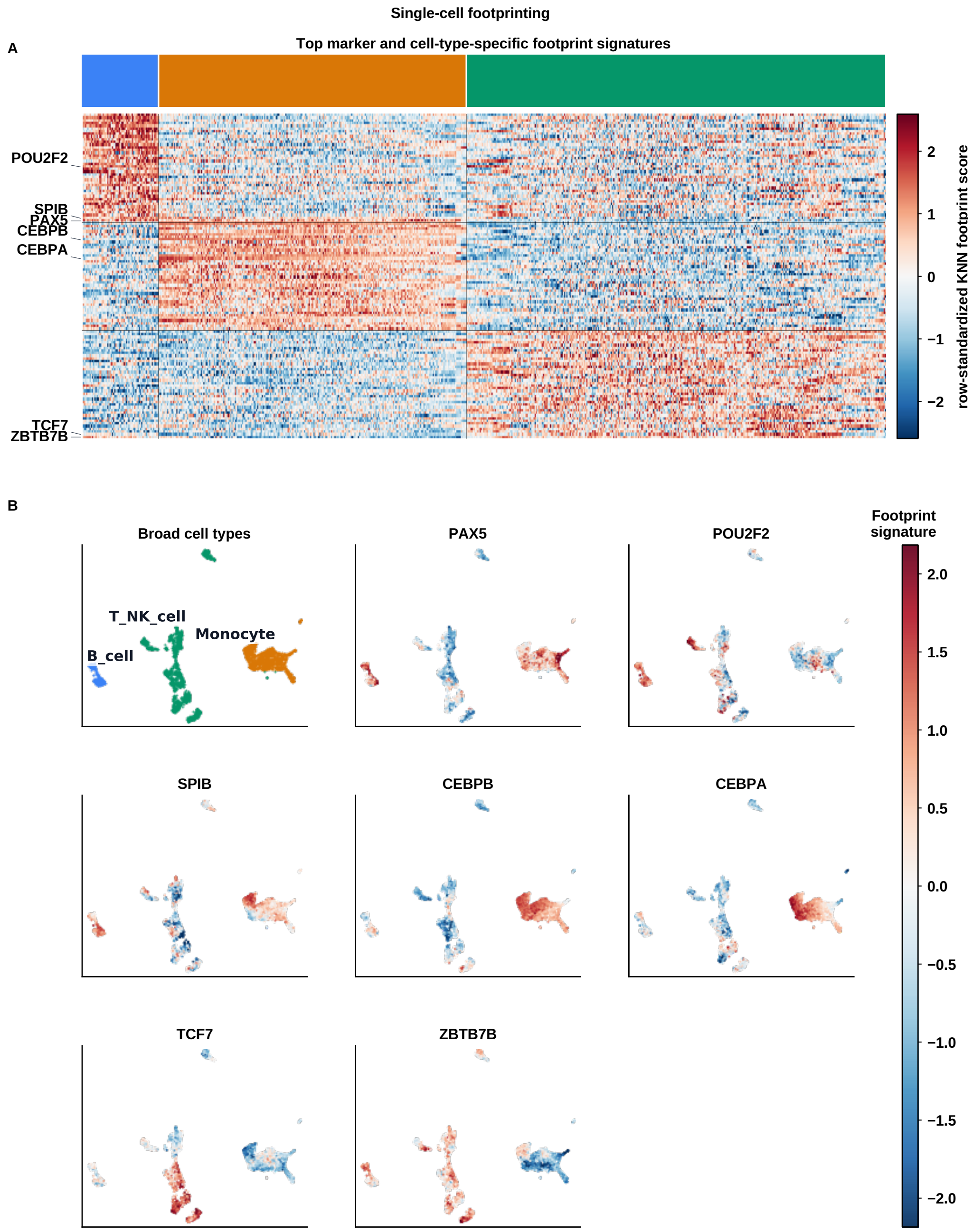

Figure 4. Single-cell footprinting summary in public PBMC5k single-cell ATAC data. (**A**) KNN-smoothed per-cell footprint-signature heatmap for selected marker TFs and top cell-type-specific motif signatures across B cells, monocytes, and T/NK cells. (**B**) UMAP projections show broad cell-type annotations and representative marker footprint-signature z-scores. Positive values indicate the marker-oriented footprint-signature direction after per-TF standardization; formal corrected footprint inference is performed at the pseudobulk level.

ATAC footprinting workflow for bias correction, footprint scoring, motif-aware differential analysis, and aggregate visualization. `fp-tools` does not claim a new Tn5 bias-correction model or a fundamentally different center-versus-flank footprint statistic. The improvement is instead at the platform layer: consistent modern command names, reusable YAML execution, report and SVG export, explicit replicate-reporting modes, de novo motif-preparation handoff, pseudobulk single-cell aggregation, GUI access, and regression-tested example outputs.

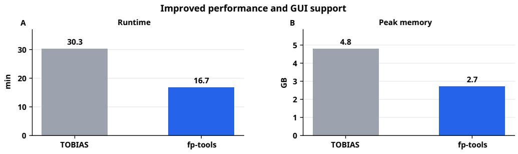

Figure 5. Representative local engineering comparison. (**A**) Runtime and (**B**) peak memory are summarized for one local workflow run comparing TOBIAS and `fp-tools` on the same analysis bundle. These measurements are included as an engineering check, not as an exhaustive cross-tool benchmark.

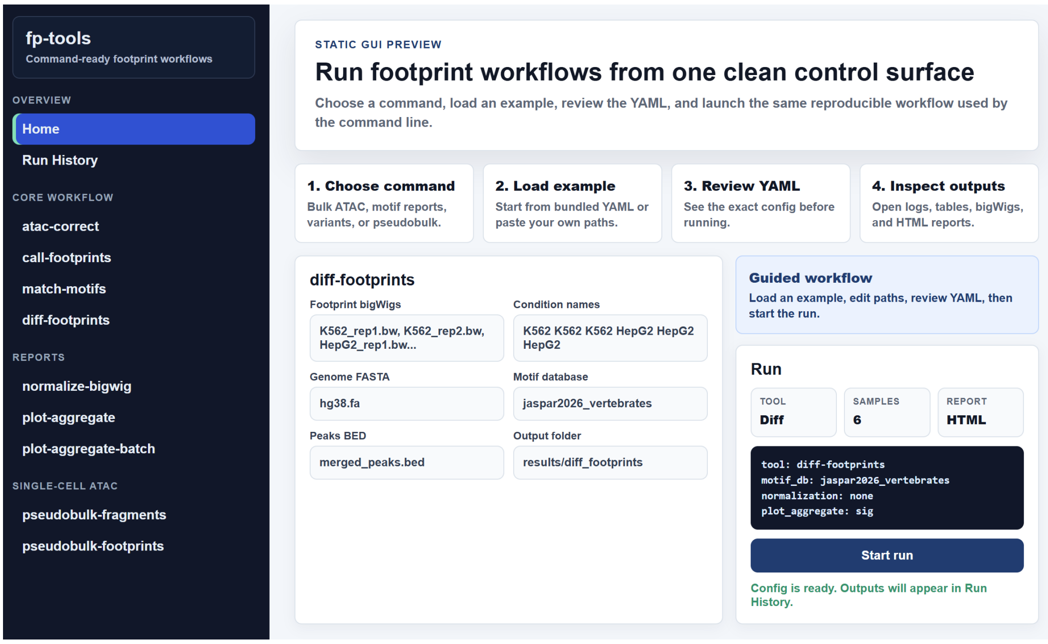

Figure 6. Browser GUI for command-compatible footprinting workflows. The GUI provides sidebar access to core commands, guided configuration panels, example loading, YAML preview, run-history review, and report/output inspection while preserving CLI-compatible configuration files.

Table 2. Qualitative scope comparison across widely used ATAC-seq regulatory-analysis tools. ✓, documented native support; ○, partial, indirect, or adjacent support; NA, outside the tool's primary scope. The table summarizes workflow scope from public documentation and primary publications; it is not a performance benchmark and should be read with the different goals of these tools in mind. TOBIAS is the closest core footprinting comparator; the `fp-tools` columns emphasize workflow extensions around that motif-centered model rather than a claim of superior footprint biology.

Feature	fp-tools	TOBIAS	HINT-ATAC	PRINT/ scPrinter	Chrom BPNet	max ATAC
Bulk ATAC footprinting	✓	✓	✓	✓	○	NA
Tn5 bias correction	✓	✓	✓	✓	✓	NA
Footprint calling	✓	✓	✓	✓	NA	NA
Motif-aware differential analysis	✓	✓	○	○	NA	NA
De novo motif preparation	✓	NA	NA	✓	○	NA
scATAC/pseudobulk support	✓	○	○	✓	○	○
Replicate-aware reporting	✓	○	○	○	NA	NA
Visualization/reporting	✓	✓	○	✓	✓	○
YAML/batch execution	✓	NA	NA	NA	NA	NA

Deep sequence-to-profile models such as ChromBPNet provide powerful sequence-driven occupancy and variant-effect analyses [10]. Differential TF activity was also considered relative to orthogonal accessibility-based frameworks such as diffTF [35]. In this broader landscape, `fp-tools` is best viewed as a practical extension of motif-centered ATAC footprinting rather than a replacement for specialized single-cell, deep-learning, or occupancy-prediction methods. The replicate-aware summaries reported here use condition-level and replicate-level score variation to aid interpretation of differential footprint comparisons. The default distribution-comparison report is available for single-sample or multi-sample condition groups, and the empirical-Bayes matrix view provides a limma-like moderated replicate analysis when a per-replicate motif-score matrix is available.

4.1. Threats to Validity and Scope

The present manuscript emphasizes reproducible workflow demonstrations, local engineering checks, and public-data examples; it is not a comprehensive cross-tool benchmark. Larger whole-genome transfer studies, including training on some cell lines and testing on others, and direct head-to-head runs of external tools such as TOBIAS, HINT-ATAC, PRINT/scPrinter, maxATAC, and ChromBPNet on identical locus sets will require larger controlled benchmark runs with repeated measurements, matched parameters, and explicit uncertainty. The runtime and memory measurements in Figure 5 are representative local engineering checks, not general performance claims.

Several biological panels are illustrative by design. The de novo recovery summary uses a merged-window overlap metric rather than an independent binding truth set, and the restricted-database sensitivity analysis depends on a specified motif-family removal list. The `highlighted` column is a report selection flag and should not be interpreted as statistical significance. The PBMC marker labels and aggregate examples emphasize expected-direction markers to show whether outputs are biologically plausible; full ranked tables are needed for formal marker-discovery claims. Finally, KNN-smoothed single-cell footprint-signature scores are visualization summaries over neighboring cells, whereas formal bias-corrected footprint inference is performed on grouped pseudobulk data.

Large public-benchmark inputs depend on external data acquisition and are therefore regenerated from committed manifests rather than stored directly in the repository. Natural next steps include larger external-tool comparisons, additional public validation datasets, expanded pseudobulk analyses across donors and tissues, and richer hierarchical models for experiments with larger replicate counts.

5. Conclusions

We presented `fp-tools`, a reproducible platform that unifies ATAC-seq footprinting with motif-discovery preparation, pseudobulk aggregation, and replicate-aware reporting. By pairing test-covered workflows with public-data manifests, source tables, and reproducible figure generation, the package supports standardized and auditable regulatory-genomics analysis. Future work will add larger whole-genome transfer studies, direct external-tool comparisons, multiscale and nucleosome-aware validation, and stronger hierarchical differential-binding validation.

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Informed Consent Statement: Not applicable.

Data Availability Statement: `fp-tools` is openly available in the public GitHub repository `oncologylab/fp-tools`; the source state for this manuscript revision is identified by the Git tag `manuscript-revision-2026-06-22`. The repository contains the benchmark manifests, schemas, analysis scripts, figure generators, source tables, and small result tables used in this manuscript. Public input data are obtained from the ENCODE Project [22], the 10x Genomics PBMC Multiome example [30], the 10x Genomics PBMC5k single-cell ATAC example [34], and motif databases such as JASPAR [14] using the included download and preparation scripts. Reviewer reproduction instructions are provided in `docs/reproduce-paper.md`; the computational environment is documented in `environment.yml` and the repository Dockerfile. Full raw public inputs are not stored in version control and are regenerated from committed manifests.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix

Table S1. Primary analysis parameters used for the public ATAC and pseudobulk workflows. The complete source table is provided in the Supplementary Materials as "Primary analysis parameters".

Step	Parameters
Public input data	Bulk ATAC analyses use Buenrostro B-cell/T-cell FASTQs and ENCODE K562/HepG2 replicate tracks. Pseudobulk examples use public granulocyte-depleted 10x PBMC Multiome and PBMC5k scATAC fragment datasets. Motif analyses use JASPAR 2026 CORE vertebrate non-redundant motifs.
FASTQ preprocessing	Paired-end adapter detection and trimming with `fastp`; HTML and JSON QC reports written for each sample.
Alignment and BAM processing	Reads aligned to hg38 with `bowtie2 -very-sensitive -X 2000`; proper pairs retained with `samtools view -f 2 -F 2828 -q 30`; BAMs sorted, fixed with `fixmate -m`, duplicate-marked with `markdup -r`, indexed, and summarized with flagstat/idxstats.
Blacklist and peak set	Mitochondrial and nonstandard contigs excluded; hg38 blacklist v2 regions subtracted; MACS3 paired-end peaks called at q-value 0.01 and merged across replicates into one analysis peak set.
Bias correction and footprint scores	`atac-correct` run on filtered BAMs with merged peaks and blacklist regions using default Tn5-bias settings; `call-footprints -score footprint` run on corrected cut-site bigWigs with default footprint and flank windows.
Differential footprint analysis	Buenrostro and ENCODE replicate footprint tracks analyzed with repeated condition labels, JASPAR 2026 motifs, and replicate reporting enabled. Runs include no-normalization and sample-quantile-normalized distribution-comparison settings where indicated; replicate-supported motif-score matrix reviews additionally use the limma-like empirical-Bayes view.
De novo motif validation	ENCODE K562/HepG2 candidate footprints exported as condition-level FASTA files with $\pm$75 bp flanks; reciprocal STREME 5.5.9 runs used `-dna -nmotifs 250`; Tomtom 5.5.9 matched discovered motifs to JASPAR 2026; de novo motifs were evaluated alone and as additions to known motif sets.
Pseudobulk PBMC workflow	10x PBMC fragments grouped by broad immune labels, retaining groups with at least 300 cells and 50,000 fragments. The full `pseudobulk-footprints` route writes pseudo-BAMs, runs ATAC correction with zero read shifting for fragment-derived pseudobulk BAMs, and emits footprint-score tracks, motif-aware differential results, and aggregate plots. The PBMC5k demonstration uses broad B-cell/monocyte/T–NK labels, marker sites, pairwise volcano plots, and KNN-smoothed footprint-signature UMAPs.

Table S2. Software and resource versions used for manuscript analyses. The complete source table is provided in the Supplementary Materials as "Software and resource versions".

Tool or resource	Version	Role
`bedtools`	2.31.1	Region sorting, merging, and blacklist subtraction
Biopython	1.87	Sequence and motif-related utilities
`bowtie2`	2.5.5	hg38 read alignment
Cell Ranger ARC	2.0.0	Source processing for the public 10x PBMC Multiome dataset
Cell Ranger ATAC	2.0.0	Source processing for the public 10x PBMC5k scATAC dataset
`fastp`	1.3.3	Paired-end FASTQ adapter detection, trimming, and QC
`fp-tools`	0.1.7	ATAC correction, footprint calling, motif matching, differential footprint analysis, pseudobulk utilities, and plotting helpers
hg38 blacklist	v2	Artifact-region filtering for processed bulk ATAC alignments
hg38 reference genome	GRCh38/hg38	Read alignment and sequence extraction
htslib	1.23.1	SAM/BAM/VCF I/O backend used by `samtools`
JASPAR CORE vertebrates	2026 non-redundant	Known motif database for differential footprint analysis
MACS3	3.0.4	ATAC-seq peak calling in paired-end mode
matplotlib	3.10.9	Manuscript figure rendering
MEME Suite, STREME, and Tomtom	5.5.9	De novo motif discovery and discovered-motif matching
NumPy	2.4.6	Numerical arrays and benchmark/figure calculations
pandas	2.3.3	Tabular processing for benchmark, manifest, and figure builders
pyBigWig	0.3.25	bigWig signal reading and writing
pysam	0.23.3	BAM and genomic interval I/O
Python	3.12.3	Runtime for analyses, figure builders, and provenance extraction
`samtools`	1.23.1	BAM filtering, sorting, fixmate, duplicate marking, indexing, and QC summaries
SciPy	1.17.1	Statistical and interpolation utilities

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